



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2002

MATHEMATICS

EXTENSION I

*Time Allowed – 2 Hours
(Plus 5 minutes reading time)*

All questions may be attempted

All questions are of equal value

In every question, show all necessary working

Marks may not be awarded for careless or badly arranged work

**Standard integral tables are included with the examination paper.
Approved silent calculators may be used.**

The answers to all questions are to be returned in separate bundles clearly labelled Question 1, Question 2, etc. Each bundle must show your candidate number.

Question 1:

- (a) Find the acute angle between the lines

$$2x + y = 17 \text{ and } 3x - y = 3$$

2

- (b) Differentiate $y = \tan^{-1} \sqrt{2x^2 - 1}$

3

- (c) Evaluate $\int_0^3 \frac{y}{\sqrt{y+1}} dy$, using the substitution $y = u^2 - 1$

3

- (d) Eight identical coins show 3 heads and 5 tails.

- (i) In how many ways can they be arranged in a straight line?

1

- (ii) What is the probability that all the tails will be together?

1

- (e) Solve for x : $\frac{2x-3}{x-2} \geq 1$

2

Question 2: (START A NEW PAGE)

- (a)

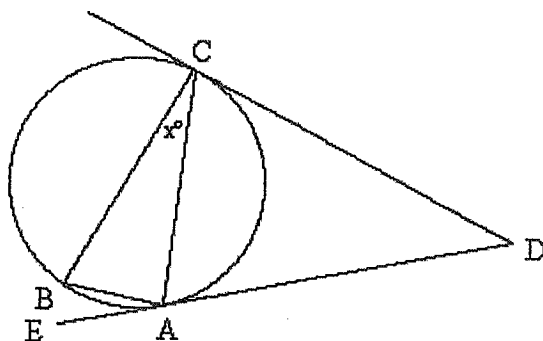


Diagram not to scale

4

AD and CD are tangents to a circle.

B is a point on the circle such that

$\angle CBA$ and $\angle CDA$ are equal and are each both double $\angle BCA$. Prove that BC

is a diameter of the circle.

- (b) The roots of the equation $9x^2 + 6x + 1 = 4kx$ where k is a real constant,

are α and β . Show that the equation with roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is

4

$$x^2 + 6x + 9 = 4kx$$

- (c) Prove by Mathematical Induction that

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$$

4

for all integers $n \geq 1$.

Question 3: (START A NEW PAGE)

- (a) The angle of elevation of a tower PQ of height h metres ^{from} at a point A due east of it is 15° . From another point B , the bearing of the tower is 032°T and the angle of elevation is 13° . The points A and B are 500 metres apart and on the same level as the base Q of the tower.
- (i) Draw a neat sketch showing all the information on your diagram 1
 - (ii) Show that $\angle AQB = 122^\circ$. 1
 - (iii) Calculate the height of the tower PQ to the nearest metre. 2
- (b) The speed v m/s of a particle moving in a straight line is given by
- $$v^2 = 64 - 16x - 8x^2$$
- where the displacement from a fixed point O is x metres.
- (i) Find an expression for the acceleration and show the motion is simple harmonic. 2
 - (ii) Find the period of the motion 1
 - (iii) Find the amplitude of the motion 1
- (c) (i) Find the largest possible domain for which
- $$f(x) = \sin^{-1}(2x+1) \text{ defines a function} \quad 1$$
- (ii) Hence find and sketch $f^{-1}(x)$, stating its domain and range. 3

Question 4: (START A NEW PAGE)

- (a) N is the number of kangaroos in a certain population at time t years.
The population size N satisfies the equation

$$\frac{dN}{dt} = -k(N - 500), \text{ for some constant } k.$$

- (i) Verify that $N = 500 + Ae^{-kt}$ with A constant, is a solution of the equation 1
- (ii) Initially, there are 3500 kangaroos but after 3 years there are only 3300 left. Find the values of A and k . 2
- (iii) Find when the number of kangaroos begins to fall below 2300 2
- (iv) Sketch the graph of the population size against time 2
- (b) An urn contains 6 cards numbered 1, 2, 3, 4, 5, 6. One card is drawn at random and a second card is drawn without the first card being replaced. Find the probability that: -
- (i) the second number is 3 1
- (ii) the larger number is 5 2
- (iii) the larger number is even 2

Question 5: (START A NEW PAGE)

- (a) At an air show, a Harrier Jump Jet leaves the ground 200 metres from an observer and rises vertically at the rate of 25 m/sec. At what rate is the observer's angle of elevation of the aircraft changing when the jet is 500 metres above the ground? 3

Question 5 continued over page.....

- (b) A chord joining the points $P(2p, p^2)$ and $Q(2q, q^2)$ on the parabola $x^2 = 4y$ passes through the point $(0, -1)$
- (i) Find the coordinates of M , the midpoint of PQ , as a function of m , the gradient of the chord 3
- (ii) Show that the cartesian equation of the locus of M is $x^2 = 2(y+1)$ for $|x| \geq 2$. 2
- (c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $A \cos(x + \alpha)$. 2
- (iii) Hence solve $\sin x + \sqrt{3} \cos x = 1$ for $0 \leq x \leq 2\pi$. 2

Question 6: (START A NEW PAGE)

- (a) The deck of a ship was 1.4 m below the level of a wharf at low tide and 0.6 m above wharf level at high tide. Low tide was at 8:24 am and hightide at 2:40pm. If tide's motion is simple harmonic, find the first time after low tide that the deck was level with the wharf. 4
- (b) Steven borrows \$50 000 to pay for a new car. He plans to repay the loan by making 60 equal monthly instalments. Interest is charged at the rate of 0.6% per month on the balance owing.
- (i) Show that immediately after making two monthly instalments of $\$P$, the balance owing is given by $\$(50\,601.80 - 2 \cdot 006P)$ 2
- (ii) Calculate the value of each monthly instalment 2
- (c) A particle is projected with an initial velocity of 60 m/s at an angle of 45° to the horizontal. (use $g = 10\text{ ms}^{-2}$)
- (i) Calculate the greatest height reached by the particle. 3
- (ii) What is the speed of the particle at the greatest height? 1

Question 7: (START A NEW PAGE)

- (a) In a box, there are 10 black counters (each marked with the digit “2”) and 5 white counters (each marked with digits “3”). 4 counters are withdrawn one at a time, the first being replaced before the second is drawn. Find the probability that
- (i) 2 blacks and 2 white counters are drawn in any order 2
 - (ii) The sum of digits on the counters drawn is greater than 9 3
- (b) (i) Show that $(1+x)^m(1-\frac{1}{x})^m = (x-\frac{1}{x})^m$ 1
- (ii) By considering the term(s) independent of x in the expansion of the result from part (b) (i), justify the result: 3

$$\binom{2002}{0} - \binom{2002}{1} + \binom{2002}{2} - \dots + \binom{2002}{2002} = -1 \binom{2002}{1001}$$

- (iii) Hence, or otherwise, show that: 3

$$\sum_{k=0}^{1001} (-1)^k \binom{2002}{k} = -\frac{1}{2} \binom{2002}{1001} \left[1 + \binom{2002}{1001} \right]$$

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

TRIAL H.S.C. EXTENSION* 2002

Q1:

(a) $2x + y = 17$ $m_1 = -2$
 $3x - y = 3$ $m_2 = 3$
 $\tan \theta = \left| \frac{-2-3}{1+2 \times 3} \right|$
 $= 1$
 $\therefore \theta = 45^\circ$ (2)

(b) $y = \tan^{-1} \sqrt{2x-1}$

$\frac{dy}{dx} = \frac{1}{\sec^2(\sqrt{2x-1})} \times \frac{1}{2\sqrt{2x-1}} \times \frac{4x}{2\sqrt{2x-1}}$
 $= \frac{1}{2x^2} \times \frac{2x}{\sqrt{2x-1}}$
 $= \frac{1}{2\sqrt{2x-1}}$ (3)

(c) $\int_0^3 \frac{y}{y+1} dy = \int_0^2 \frac{u^2-1}{u} \cdot 2u du$ $y = u^2-1$
 $= \int_0^2 (2u^2-2) du$ $\frac{dy}{du} = 2u$
 $= \left[\frac{2u^3}{3} - 2u \right]_0^2$ $dy = 2u du$
 $= \frac{16}{3} - 4$ $\text{When } y=0, u=1$
 $= \frac{4}{3}$ $\text{When } y=3, u=2$ (3)

(d) (i) no. of ways = $\frac{8!}{5!3!} = 56$ (1)
(ii) P(all tails tog.) = $\frac{4}{56} = \frac{1}{14}$

TTTTTHHH
HTTTTTHH (1)
HHTTTTTT
HHHTTTTT

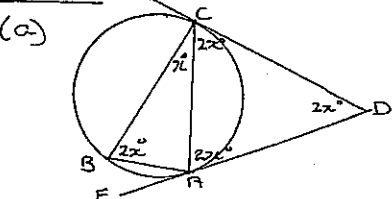
(e) $\frac{2x-3}{x-2} \geq 1$

$\frac{2x-3-x-2}{x-2} \geq 0$
 $\frac{x-5}{x-2} \geq 0$

+	0	+
-	+	$x-1$
-	-	$x-2$
		2+

$\therefore x \leq 1$ and $x > 2$ (2)

Q2:



$\angle CBA = \angle CDA = 2x^\circ$ (given)

$\angle DAC = \angle CBA$ (angle between a tangent & a chord equals angle in the alternate segment)

Similarly

$\angle DCA = \angle CBA = 2x^\circ$ (1)

$\therefore$ In $\triangle CDA$ $2x + 2x + 2x = 180^\circ$ (angle sum $\triangle$)

$\therefore x = 30^\circ$ (1)

In $\triangle BAC$

$x^\circ + 2x^\circ + \angle BAC = 180^\circ$

$30^\circ + 60^\circ + \angle BAC = 180^\circ$

$\therefore \angle BAC = 90^\circ$ (1)

$\therefore BC$ is a diameter (angle in semi-circle is 90°) (1)

(b) $9x^2 + 6x + 1 = 4kx$

$9x^2 + (6-4k)x + 1 = 0$

$\alpha + \beta = \frac{4k-6}{9}$ (1)

$\alpha\beta = \frac{1}{9}$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{4k-6}{1} = \frac{-b}{a}$ (1)

$\frac{1}{\alpha} \times \frac{1}{\beta} = \frac{1}{9} = \frac{c}{a}$

$\therefore a = 1$
 $b = 6-4k$ (1)
 $c = 9$

$\therefore$ Eqn. is $x^2 + (6-4k)x + 9 = 0$ (1)
 $9x^2 + 6x + 9 = 4kx$

(c) Let $P(n)$ be proposition

$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n$

Step 1: For $P(1)$

$1 \times 2^0 = 1 + (1-1)2^1$
 $1 = 1$ (1)
 $\therefore P(1)$ is true

Step 2: Assume that $P(k)$ is true for some integer $k \geq 1$

i.e. $P(k): 1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} = 1 + (k-1)2^k$ (1)
and RTS $P(k+1)$ is true

Proof: For $P(k+1)$

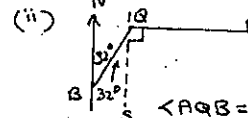
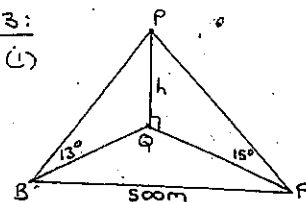
$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + k \times 2^{k-1} + (k+1)2^k$
 $= 1 + (k-1)2^k + (k+1)2^k$
 $= 1 + ((k-1) + (k+1))2^k$
 $= 1 + 2k \cdot 2^k$
 $= 1 + k \cdot 2^{k+1}$ (2)

$\therefore P(k+1)$ is true

Step 3: If the result is true for $P(1)$, assumed true for $P(k)$ and proven true for $P(k+1)$ then it is true for all positive integral values of n .

Q3:

(a) (i)



$\angle AQB = 122^\circ$ as $QA \perp QS$ & $\angle BQS = 32^\circ$ (1)

(iii) In $\triangle APQ$

$h = AQ \cot 15^\circ$

In $\triangle PQB$

$h = BQ \cot 13^\circ$

In $\triangle ABQ$

$500^2 = h^2 \cot^2 13^\circ + h^2 \cot^2 15^\circ - 2 \cdot h \cdot h \cot 13^\circ \cot 15^\circ \cos 122^\circ$

$h^2 = \frac{500^2}{\cot^2 13^\circ + \cot^2 15^\circ + 2 \cot 13^\circ \cot 15^\circ \cos 58^\circ}$

$h = \frac{500}{\sqrt{\cot^2 13^\circ + \cot^2 15^\circ + 2 \cot 13^\circ \cot 15^\circ \cos 58^\circ}}$

$\therefore h = 71 \text{ m}$ (nearest m) (2)

(b) $V^2 = 64 - 16x - 8x^2$

(i) $\frac{1}{2} V^2 = 32 - 8x - 4x^2$

$\frac{d}{dx}(\frac{1}{2} V^2) = -8 - 8x$

$\therefore \dot{x} = -8(x+1)$

$\therefore$ Motion is SHM centre at $x = -1$ (2)

(ii) $P_{\text{period}} = \frac{2\pi}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}} \text{ sec}$ as $n = 2\sqrt{2}$ (1)

(iii) For motion to exist $v^2 \geq 0$

$8(4+x)(2-x) \geq 0$

$-4 \leq x \leq 2$ $\frac{1}{4} \quad -1 \quad 2$ (1)

$\therefore$ Amplitude = 3m

(c) (i) D: $-1 \leq 2x+1 \leq 1$

$-2 \leq 2x \leq 0$

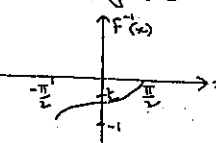
$-1 \leq x \leq 0$ (1)

(ii) $x = \sin^{-1}(2y+1)$

$y = \frac{1}{2}(\sin x - 1)$

D_{P-1} = $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

R_{P-1} = $-1 \leq y \leq 0$ (2)



Q4:

(a) (i) $N = 500 + Ae^{-kt}$

$\frac{dN}{dt} = -kAe^{-kt}$

$\therefore \frac{dN}{dt} = -k(N-500)$ as $Ae^{-kt} = N-500$

(ii) $A = 3000$ when $t = 0$

When $t = 3$ $N = 3300$

$3300 = 500 + 3000e^{-3k}$

$\ln \frac{14}{15} = -3k$

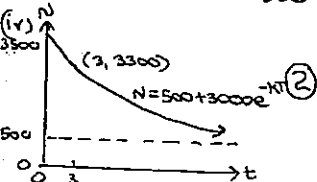
$k = \frac{1}{3} \ln \frac{14}{15}$

(iii) $500 + 3000e^{-kt} \leq 2300$

$e^{-kt} \leq \frac{1800}{3000}$

$\therefore t \geq 22.21 \text{ yrs}$

$\therefore$ No. falls below 2300 when $t \geq 22$



(b) (i) Sample space = ${}^6P_2 = 30$

No. of favourable events = 5

i.e. (1,3) (3,3) (4,3) (5,3) (6,3)

$P(\text{2nd. no. is 3}) = \frac{5}{30} = \frac{1}{6}$ (1)

(ii) Sample space = ${}^6C_2 = 15$

No. of favourable events = 4

$P(\text{larger no. is a 5}) = \frac{4}{15}$ (2)

(iii) $n(S) = {}^6C_2 = 15$

5 has 2 larger even no's.

3 " 4 " " "

1 " 6 " " "

$\therefore n(E) = 9$

$P(\text{larger no. even}) = \frac{9}{15} = \frac{3}{5}$ (2)

7 (a) 10 blocks ("2") 5 white ("3")

$$P(0) = P(2) = \frac{10}{15} = \frac{2}{3}$$

$$P(1) = P(3) = \frac{1}{3}$$

(b) $P(2 \text{ or } 3) = 4! \cdot P(0) \cdot P(4) = 4! \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2$

$$= \frac{24}{27} = \frac{8}{9} = 0.888$$

(ii) B(4) W(3) Total P(5)

$$\left(\frac{2}{3} + \frac{1}{3}\right)^4$$

$$4 \quad 0 \quad 1 \quad \left(\frac{2}{3}\right)^4 = \frac{16}{81}$$

$$3 \quad 1 \quad 7 \quad 4 \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right) = \frac{32}{81}$$

$$2 \quad 2 \quad 6 \quad 6 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2 = \frac{24}{81}$$

$$1 \quad 3 \quad 4 \quad 4 \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^3 = \frac{8}{81}$$

$$0 \quad 4 \quad 1 \quad \left(\frac{1}{3}\right)^4 = \frac{1}{81}$$

$$P(X > 4) = P(2 \text{ or } 3 \text{ or } 4 \text{ or } 5)$$

$$= P(2) + 4 \cdot \frac{2}{3} \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^4 = \frac{22}{81} = \frac{11}{27}$$

$$= 1 - P(X \leq 4)$$

$$= 1 - P(4 \text{ or } 5 \text{ or } 6 \text{ or } 7)$$

$$1 - \frac{48}{81} = 1 - \frac{16}{27} = \frac{11}{27} = 0.407$$

$$(c) (1+x)^n (1-\frac{1}{x})^n = \left[(1+x) \left(1-\frac{1}{x}\right) \right]^n$$

$$= \left[1 - \frac{1}{x} + x - \frac{1}{x} \right]^n$$

$$= \left[x - \frac{1}{x} \right]^n$$

$$(d) LHS = (1+x)^{2002} (1-\frac{1}{x})^{2002}$$

$$= \left[\binom{2002}{0} + \binom{2002}{1}x + \binom{2002}{2}x^2 + \dots + \binom{2002}{2002}x^{2002} \right] \left[\binom{2002}{0} - \binom{2002}{1}\frac{1}{x} + \binom{2002}{2}\frac{1}{x^2} - \dots + (-1)^{2002} \left(\frac{1}{x}\right)^{2002} \right]$$

$$\text{Coeff of } x^0 = \binom{2002}{0} \binom{2002}{0} + \binom{2002}{1} \binom{2002}{1} (-1) + \binom{2002}{2} \binom{2002}{2} (-1)^2 + \dots + \binom{2002}{2002} \binom{2002}{2002} (-1)^{2002}$$

match if they add up until last term when they subtract - need to have $(-1)^{2002}$

$$RHS = (x - \frac{1}{x})^{2002} = (x - \frac{1}{x})^{2002}$$

$$\text{Total} = \sum_{k=0}^{2002} \binom{2002}{k} x^{k-2002} = \sum_{k=0}^{2002} \binom{2002}{k} (-1)^{2002-k} x^{k-2002}$$

$$x^k = x^{k-2002} \Rightarrow k-2002 = 0 \Rightarrow k = 2002 \quad (x = 1001)$$

$$x = \binom{2002}{2002} (-1)^{0} = 1 \quad (\text{need to show not quite})$$

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APPROACH 1

$$LHS = \sum_{k=0}^{2002} \binom{2002}{k} (-1)^k = \left(\binom{2002}{0} - \binom{2002}{1} + \binom{2002}{2} - \dots + (-1)^{2002} \binom{2002}{2002} \right)$$

$$= \sum_{k=0}^{2002} \binom{2002}{k} (-1)^k = \left(\sum_{k=0}^{2002} \binom{2002}{k} (-1)^k \right)$$

$$\text{Let } \left(\frac{2002}{0} - \frac{2002}{1} + \frac{2002}{2} - \dots + \frac{2002}{2002} \right) = \left(\frac{2002}{0} - \frac{2002}{1} + \frac{2002}{2} - \dots + \frac{2002}{2002} \right)$$

$$\sum_{k=0}^{2002} \binom{2002}{k} (-1)^k = \left(\sum_{k=0}^{2002} \binom{2002}{k} (-1)^k \right)$$

$$= -1 \left(\frac{2002}{0} - \frac{2002}{1} + \frac{2002}{2} - \dots + \frac{2002}{2002} \right)$$

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$$= -1 \left(\frac{2002}{0} - \frac{2002}{1} + \frac{2002}{2} - \dots + \frac{2002}{2002} \right)$$

$$2 \sum = -1 \left(\frac{2002}{0} - \frac{2002}{1} + \frac{2002}{2} - \dots + \frac{2002}{2002} \right)$$

$$\Rightarrow \sum = -\frac{1}{2} \left(\frac{2002}{0} - \frac{2002}{1} + \frac{2002}{2} - \dots + \frac{2002}{2002} \right)$$

APPROACH II

$$LHS = \sum_{k=0}^{2002} \binom{2002}{k} (-1)^k = \left(\binom{2002}{0} - \binom{2002}{1} + \binom{2002}{2} - \dots + \binom{2002}{2002} (-1)^{2002} \right)$$

$$\sum_{k=0}^{2002} \binom{2002}{k} (-1)^k = \left(\binom{2002}{0} - \binom{2002}{1} + \binom{2002}{2} - \dots + \binom{2002}{2002} (-1)^{2002} \right)$$

$$\sum_{k=0}^{2002} \binom{2002}{k} (-1)^k = \left(\binom{2002}{0} - \binom{2002}{1} + \binom{2002}{2} - \dots + \binom{2002}{2002} (-1)^{2002} \right)$$

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